

Laikipia County Job Portal

System Documentation — Recruitment Lifecycle & Architecture

County Government of Laikipia — ICT Attachment Project

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1. Introduction

The Laikipia County Job Portal is a full-stack recruitment system built for the County Government of Laikipia’s Public Service Board. It digitizes the entire hiring lifecycle — from a department first requesting to fill a post, through public advertisement, applications, shortlisting, interviews (in-person and online), panel scoring, and final appointment.

Technology stack

Layer	Technology
Backend	Java 21, Spring Boot, Spring Security (JWT), Spring Data JPA / Hibernate
Database	MySQL
Frontend	React (Vite), Tailwind CSS
Email	SMTP (Gmail) for transactional notifications
Auth	Email/password with 6-digit email verification codes; JWT bearer tokens

Design system — the frontend uses a “Heritage Ledger” visual identity (deep forest green and gold palette, Merriweather/Inter typefaces) reflecting the County Public Service Board’s institutional character.

2. User Roles

The system is built around six roles, enforced both at the API layer (@PreAuthorize) and in the frontend's route guards:

Role	Description
SUPER_ADMIN	Full system control. Creates staff accounts, manages departments, can act as a safety-net approver in every workflow, and can mark applications as done.
CPSB_ADMIN	County Public Service Board Administrator. Approves/rejects recruitment requests from departments, selects final candidates, and confirms appointments.
DEPT_HEAD	Department Head. Submits recruitment requests (demand for a vacancy) on behalf of their department.
HR_OFFICER	Human Resources Officer. Reviews and updates application statuses, manages shortlists, schedules interviews, and — together with SUPER_ADMIN — can mark applications as done.
PANEL_MEMBER	Interview panellist. Scores candidates in both traditional (in-person) and online interviews, and creates/manages the online interview question bank.
APPLICANT	Member of the public. Registers, builds a profile, applies for vacancies, uploads documents, and completes online interviews.

3. End-to-End Recruitment Flow

The system models recruitment as a strict pipeline. Each stage gates the next — you cannot skip ahead (e.g. a vacancy cannot be created until its recruitment request is approved; an application cannot be selected until it has reached the interview stage).

3.1 Recruitment Request (Demand Creation)

1. A **Department Head** submits a recruitment request for their own department, specifying the position, grade, and justification.
2. A **CPSB Admin** reviews the request and either **approves** or **rejects** it. The approver's identity is taken from their authenticated session — never from client-supplied data — so a request cannot be self-approved or approved by an impersonated user.
3. Only an **APPROVED** request can be used to create a vacancy.

3.2 Vacancy Creation & Advertisement

1. Once a recruitment request is approved, HR/Admin creates a **Job Vacancy** linked to it — title, department, description, requirements, number of positions, and an application deadline.
2. The vacancy is published with status **OPEN** and appears on the public vacancies list for anyone to browse, whether logged in or not.
3. When positions are filled (see §3.9), the vacancy automatically flips to **FILLED** and stops accepting new applications.

3.3 Applicant Onboarding

1. A member of the public **registers** with an email and password.
2. The system emails a **6-digit verification code**; the applicant must verify their email before logging in.
3. Once verified, the applicant logs in and completes their **profile**: national ID, date of birth, gender, nationality, physical address, county of birth/residence, educational level, and years of experience. This profile must be complete before any application can be submitted.

3.4 Application Submission

1. The applicant selects an open vacancy and submits an application, including a suitability statement, a set of legal declarations (no conflict of interest, no criminal conviction, consent to data processing, etc.), and **at least one referee**.
2. The system blocks duplicate applications — one applicant cannot apply to the same vacancy twice.
3. On submission, the applicant receives both an in-app notification and an email acknowledgement with a reference number.

3.5 Document Upload

1. The applicant uploads supporting documents (ID copy, certificates, testimonials, etc.) against their application.
2. Documents are strictly tied to the owning applicant — an applicant can only attach files to their own application, never to someone else's.

3.6 Shortlisting

1. An **HR Officer** or **CPSB Admin** reviews submitted applications for a vacancy and moves qualifying candidates to **SHORTLISTED**.
2. Only applications in **SUBMITTED** or **UNDER_REVIEW** state can be shortlisted, and an application cannot be shortlisted twice.

3.7 Interviews

The system supports two interview formats, which can be used independently or together depending on the vacancy:

3.7.1 Traditional (in-person / video-call) interviews

1. HR schedules an interview for a shortlisted application — date, time, venue/link — and assigns one or more **panel members**.
2. Panel members score the candidate (technical, communication, and experience scores, each 0-100) once the interview has taken place.
3. HR marks the interview **COMPLETED** once all panel scoring is done.

3.7.2 Online (asynchronous, question-set-based) interviews

1. Panel members build a reusable **question bank**: multiple-choice, true/false, checkbox, or free-text questions, each with a difficulty level and default marks.
2. Questions are grouped into a **Question Set** and published against a vacancy.
3. When HR schedules an online interview for a shortlisted applicant, the system:
 - creates the interview record with a unique, unguessable access token,
 - sends the applicant an **in-app notification and email** with a direct portal link,
 - the invite appears in the applicant's dashboard automatically — the portal polls for new interviews every few seconds and also refreshes instantly when the applicant returns to the tab, so there's no need to manually refresh to see a newly scheduled interview.
4. The applicant opens the link, starts the interview (a time window applies — open-sAt/closesAt), answers each question, and submits.
5. Panel members then review submitted answers. **The review screen shows the exact same option list, in the exact same layout, that the applicant saw when answering** — with the applicant's selection and the correct answer both highlighted — rather than a flattened text summary. This same shared component is also used when panel members preview a question set while building it,

so the question-authoring view, the applicant's answering view, and the panel's review view are all visually and structurally identical.

6. A panel member finalizes the result with a recommendation (Shortlist / Reject / Hold) once every answer has been scored by at least one panellist.

3.8 Final Selection

1. Once an application has progressed to the **INTERVIEW** stage, a **CPSB Admin** can select the candidate, moving the application to **SELECTED** and creating a Final Selection record with status **PENDING_APPOINTMENT**.
2. An applicant can only be selected once per application.

3.9 Appointment (Employment)

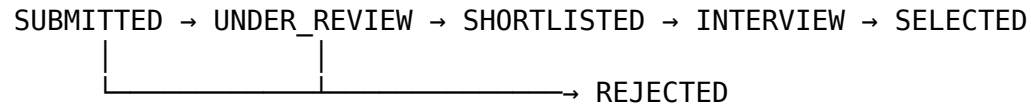
1. The CPSB Admin updates the Final Selection's appointment status to **APPOINTED** — this is the point at which a candidate is formally employed.
2. Once enough candidates have been appointed to fill all open positions on a vacancy, the vacancy automatically transitions to **FILLED** and no longer accepts new applications.

3.10 Closing Out — “Mark as Done”

Once an application has run its course (appointed, rejected, or otherwise concluded), it can clutter the working queues that HR and Admin use daily. To keep those queues focused on active work:

1. A **SUPER_ADMIN** or **HR_OFFICER** can mark any individual application as **Done**.
2. Once marked done, the application is **hidden from the default applications/shortlisting views** — it no longer appears in the day-to-day working lists.
3. It is **never deleted or hidden from the applicant** — the applicant still sees it in their own “My Applications” list at all times.
4. HR/Admin can toggle a “**Show done applications**” switch on the shortlisting workspace to see closed applications again, and can **reopen** one if it was closed by mistake.

4. Application State Machine



Independently of this status, every application also carries a **closed** flag (see §3.10) — closing an application does not change its status, it only affects whether it shows up in the default HR/Admin queues.

5. Notifications & Communication

Every meaningful transition triggers both an **in-app notification** and, for key events, an **email**:

- Application received
- Application rejected
- Online interview invitation (with a direct, clickable portal link — the notifications page auto-detects the link in the message and renders it as an actionable button rather than raw text)
- Shortlisting and interview scheduling updates

6. Data Model Overview (Key Entities)

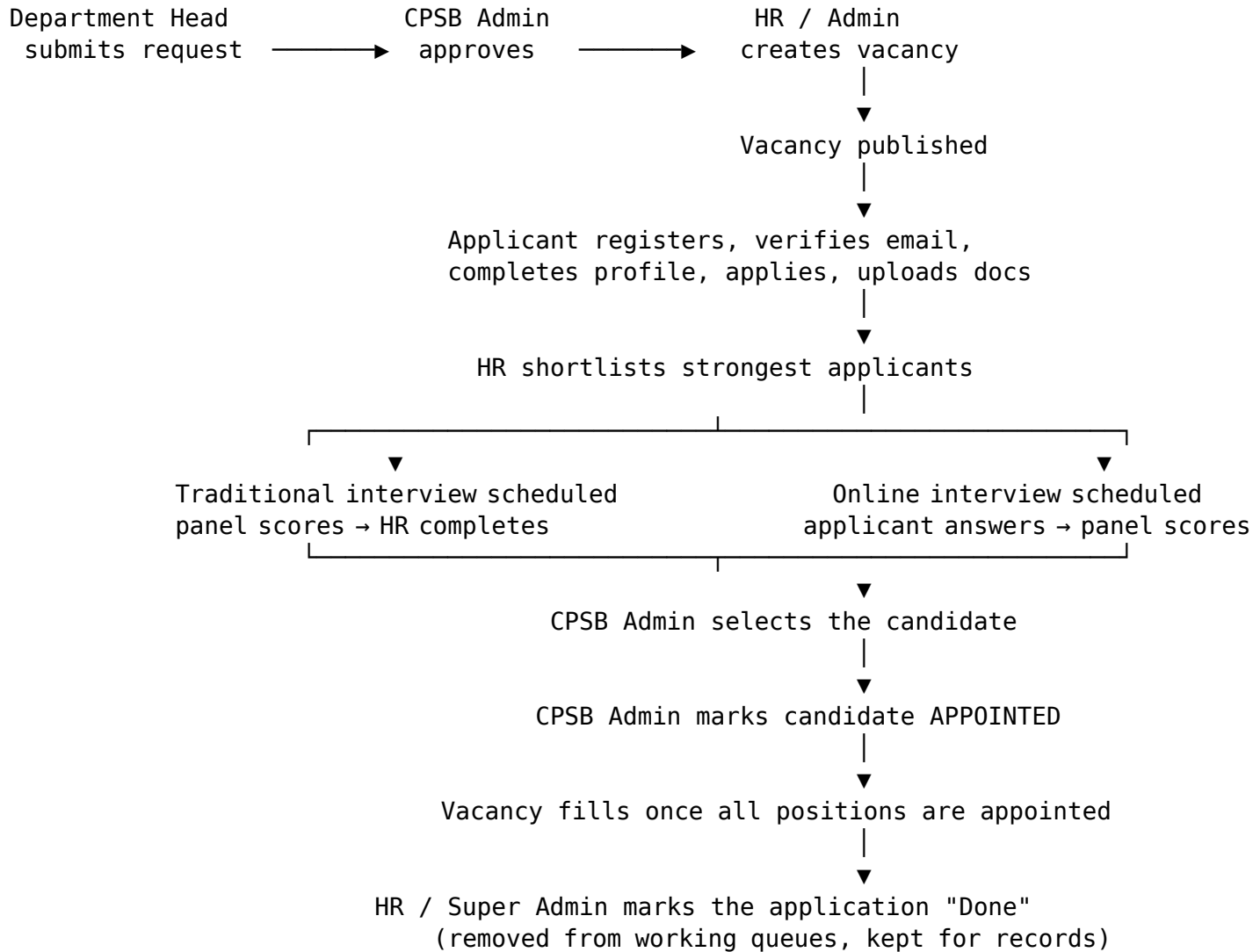
Entity	Purpose
Users	Account record for every role; holds credentials, role, and verification state.
Applicant	Extended profile data for APPLICANT users (national ID, education, experience, etc.).
Departments	County departments; each has a designated Department Head.
RecruitmentRequest	A department's demand to fill a post; requires CPSB approval.
JobVacancy	A published, applicable post, linked to an approved recruitment request.
Applications	An applicant's application to a vacancy; carries status, referees, declarations, and the closed/closedAt/closedBy "mark as done" fields.
ApplicationDocument	Uploaded supporting file, owned by one application.
Interview / InterviewPanel / InterviewScore	Traditional in-person interview scheduling, panel assignment, and scoring.
OnlineInterview	Token-based asynchronous interview instance for one application.
QuestionSet / InterviewQuestion / InterviewQuestionOption	The reusable online-interview question bank.
ApplicantAnswer / ApplicantAnswerScore	An applicant's submitted answers and the panel's scoring of them.
OnlineInterviewResult	Finalized recommendation and aggregate score for an online interview.
FinalSelection	Selection + appointment record — the bridge from "interviewed" to "employed".

7. Security & Reliability Notes

The following safeguards are enforced throughout the pipeline:

- **Authenticated-identity checks, not client-supplied IDs.** Approver, scorer, and document-owner checks are always resolved from the authenticated session (JWT), never trusted from the request body.
- **Ownership enforcement on documents.** An applicant may only upload or view documents against their own application.
- **Consistent error responses.** Not-found conditions return 404, authorization failures return 403, and business-rule violations (e.g. “already shortlisted”, “vacancy not open”) return 409 — rather than a generic 500 — so the frontend (and any API consumer) can react appropriately.
- **Ordered scoring workflow.** Panel members can score an interview once it has taken place; HR then marks it complete once scoring is done — scoring is only blocked for a cancelled interview.
- **Consistent question/answer presentation.** The same visual component renders question options everywhere they appear (authoring, answering, and reviewing), so there is no risk of the panel misreading an answer due to a different layout than the one the applicant used.

8. Summary of the Full Journey



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